DSC640 Weeks 9 & 10 Assignment - TSA ComplaintsAyden Trusty

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1.1 DSC640 Ayden Trusty

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Title: Assignment Weeks 9 & 10

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Date: 28 February 2026

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Description: This program is to demonstrate my understanding/practice of using Python to review datasets and plot visuals to tell a story.

[19]:
```
import pandas as pd
import matplotlib.pyplot as plt
```

[2]:
```
airport_df = pd.read_csv("complaints-by-airport.csv")
cat_df     = pd.read_csv("complaints-by-category.csv")
subcat_df  = pd.read_csv("complaints-by-subcategory.csv")
iata_df    = pd.read_csv("iata-icao.csv")
```

[11]:
```
airport_df.head()
```

[11]:
```
  pdf_report_date airport year_month  count
0         2019-02     ABE    2015-01      0
1         2019-02     ABE    2015-02      0
2         2019-02     ABE    2015-03      0
3         2019-02     ABE    2015-04      0
4         2019-02     ABE    2015-05      2
```

[12]:
```
cat_df.head()
```

[12]:
```
  pdf_report_date airport                           category year_month  \
0         2019-02     ABE         Hazardous Materials Safety    2015-01
1         2019-02     ABE  Mishandling of Passenger Property    2015-01
2         2019-02     ABE         Hazardous Materials Safety    2015-02
3         2019-02     ABE  Mishandling of Passenger Property    2015-02
```

```
4         2019-02     ABE         Hazardous Materials Safety    2015-03

   count                          clean_cat clean_cat_status
0      0         Hazardous Materials Safety         original
1      0  Mishandling of Passenger Property         original
2      0         Hazardous Materials Safety         original
3      0  Mishandling of Passenger Property         original
4      0         Hazardous Materials Safety         original
```

[13]:
```
subcat_df.head()
```

[13]:
```
  pdf_report_date airport                           category  \
0         2019-02     ABE         Hazardous Materials Safety
1         2019-02     ABE  Mishandling of Passenger Property
2         2019-02     ABE         Hazardous Materials Safety
3         2019-02     ABE  Mishandling of Passenger Property
4         2019-02     ABE         Hazardous Materials Safety

                               subcategory year_month  count  \
0                                  General    2015-01      0
1  Damaged/Missing Items--Checked Baggage    2015-01      0
2                                  General    2015-02      0
3  Damaged/Missing Items--Checked Baggage    2015-02      0
4                                  General    2015-03      0

                           clean_cat                            clean_subcat  \
0         Hazardous Materials Safety                                 General
1  Mishandling of Passenger Property  *Damaged/Missing Items--Checked Baggage
2         Hazardous Materials Safety                                 General
3  Mishandling of Passenger Property  *Damaged/Missing Items--Checked Baggage
4         Hazardous Materials Safety                                 General

  clean_cat_status clean_subcat_status  is_category_prefix_removed
0         original            original                       False
1         original            original                       False
2         original            original                       False
3         original            original                       False
4         original            original                       False
```

[14]:
```
iata_df.head()
```

[14]:
```
  country_code region_name iata  icao                          airport  \
0           AE    Abu Zaby  AAN  OMAL     Al Ain International Airport
1           AE    Abu Zaby  AUH  OMAA  Abu Dhabi International Airport
2           AE    Abu Zaby  AYM   NaN         Yas Island Seaplane Base
3           AE    Abu Zaby  AZI  OMAD      Al Bateen Executive Airport
4           AE    Abu Zaby  DHF  OMAM               Al Dhafra Air Base
```

```
   latitude  longitude
0   24.2617    55.6092
1   24.4330    54.6511
2   24.4670    54.6103
3   24.4283    54.4581
4   24.2482    54.5477
```

[15]:
```
def parse_year_month(s):
    # year_month looks like "YYYY-MM"
    return pd.to_datetime(s + "-01", errors="coerce")

for df in [airport_df, cat_df, subcat_df]:
    df["ym"] = parse_year_month(df["year_month"].astype(str))
```

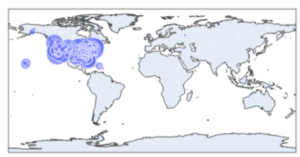
[16]:
```
# Merge airport lookup (IATA -> airport name + lat/long)
# complaints datasets use 'airport' as the IATA code
airport_lookup = iata_df.rename(columns={"iata": "airport", "airport":␣
 ↪"airport_name"})
airport_df = airport_df.merge(airport_lookup[["airport", "airport_name",␣
 ↪"latitude", "longitude"]], on="airport", how="left")
cat_df     = cat_df.merge(airport_lookup[["airport", "airport_name",␣
 ↪"latitude", "longitude"]], on="airport", how="left")
subcat_df  = subcat_df.merge(airport_lookup[["airport", "airport_name",␣
 ↪"latitude", "longitude"]], on="airport", how="left")
```

[17]:
```
airport_totals = (airport_df
                  .groupby(["airport", "airport_name", "latitude",␣
 ↪"longitude"], dropna=False)["count"]
                  .sum()
                  .reset_index()
                  .sort_values("count", ascending=False))

top_airports = airport_totals.head(15)["airport"].tolist()

# Total complaints by category (for top categories)
cat_totals = (cat_df
              .groupby(["clean_cat"], dropna=False)["count"]
              .sum()
              .reset_index()
              .sort_values("count", ascending=False))

top_cats = cat_totals.head(10)["clean_cat"].tolist()
```

[20]:
```
import plotly.express as px

map_df = airport_totals.dropna(subset=["latitude", "longitude"]).head(200)
```

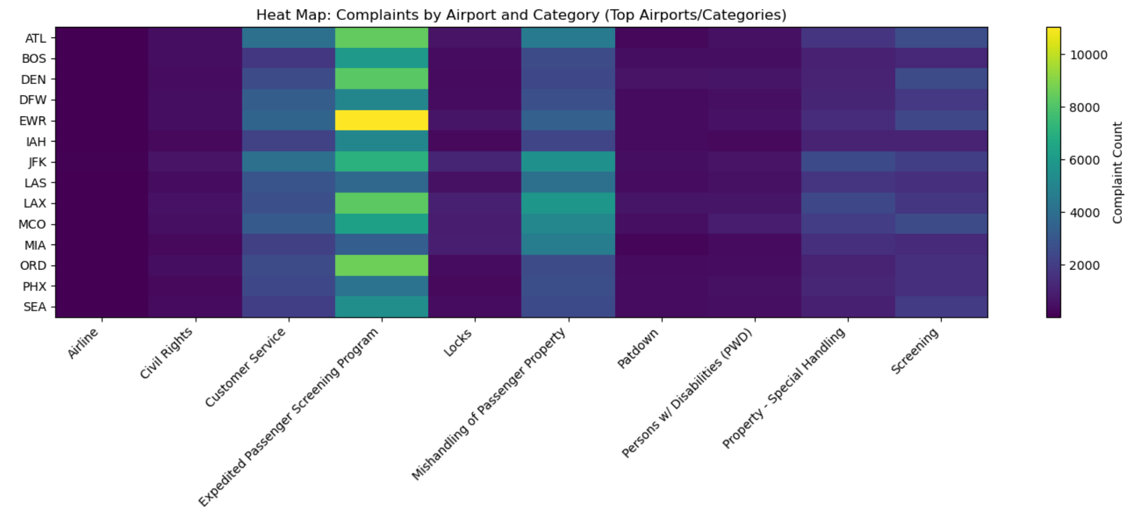
```
fig = px.scatter_geo(
    map_df,
    lat="latitude",
    lon="longitude",
    size="count",
    hover_name="airport_name",
    hover_data={"airport": True, "count": True, "latitude": False, "longitude": False},
    title="TSA Complaints Concentration by Airport (Top 200)"
)
fig.show()
```

TSA Complaints Concentration by Airport (Top 200)

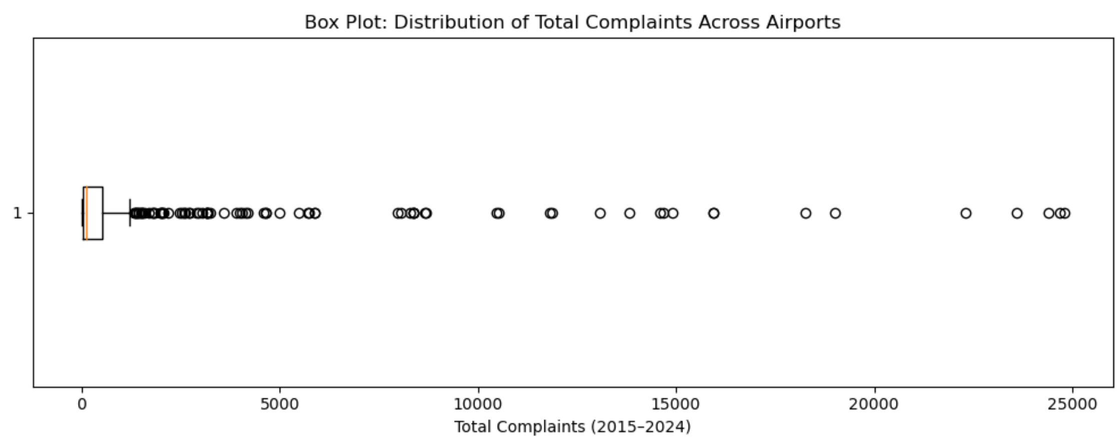


[21]:

```
heat_df = cat_df[cat_df["airport"].isin(top_airports) & cat_df["clean_cat"].isin(top_cats)].copy()

heat_pivot = (heat_df
              .groupby(["airport", "clean_cat"])["count"]
              .sum()
              .reset_index()
              .pivot(index="airport", columns="clean_cat", values="count")
              .fillna(0))

plt.figure(figsize=(14, 6))
plt.imshow(heat_pivot.values, aspect="auto")
plt.xticks(range(len(heat_pivot.columns)), heat_pivot.columns, rotation=45, ha="right")
plt.yticks(range(len(heat_pivot.index)), heat_pivot.index)
plt.title("Heat Map: Complaints by Airport and Category (Top Airports/Categories)")
plt.colorbar(label="Complaint Count")
plt.tight_layout()
plt.show()
```



[32]:
```
plt.figure(figsize=(10, 4))
plt.boxplot(airport_totals["count"], vert=False)
plt.title("Box Plot: Distribution of Total Complaints Across Airports")
plt.xlabel("Total Complaints (2015-2024)")
plt.tight_layout()
plt.show()
```



[33]:
```
airport_totals.dtypes
airport_totals[["airport","airport_name"]].head()
```

[33]:

	airport	airport_name
234	LAX	Los Angeles International Airport
220	JFK	John F. Kennedy International Airport
138	EWR	Newark Liberty International Airport

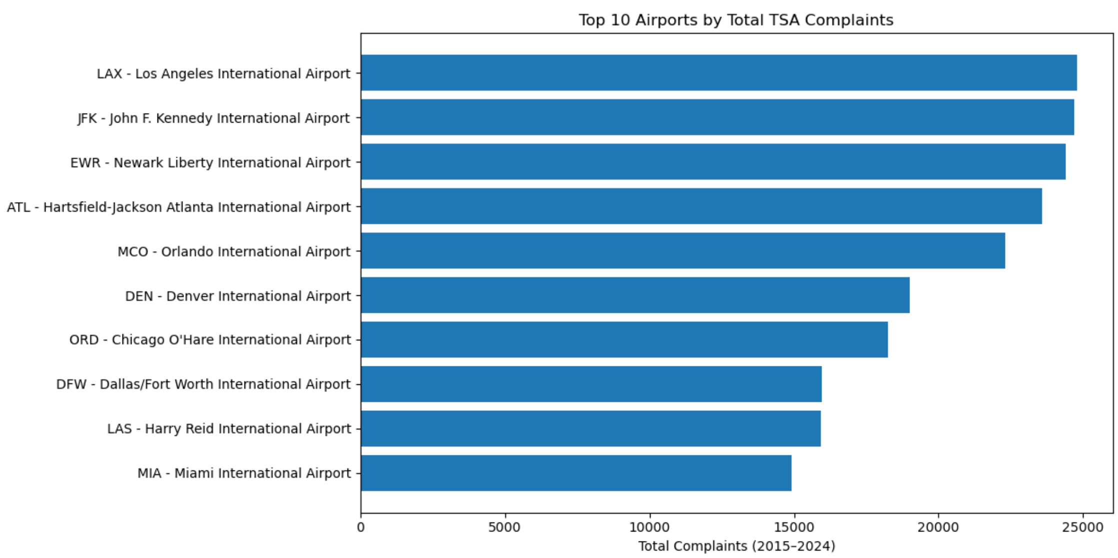
```
26       ATL  Hartsfield-Jackson Atlanta International Airport
263      MCO                         Orlando International Airport
```

[34]:
```
airport_totals = airport_totals.dropna(subset=["airport"])
```

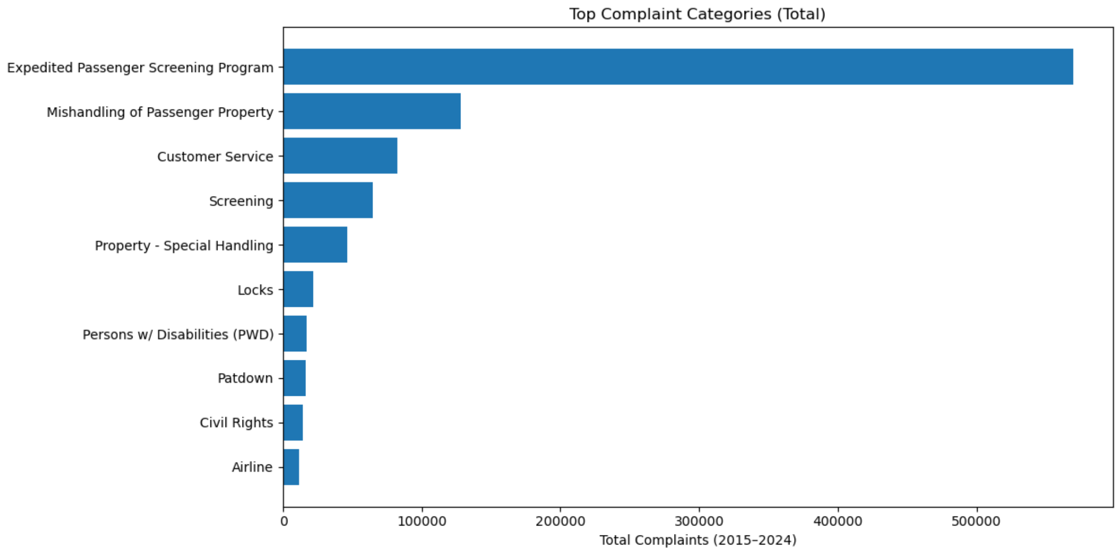
[35]:
```
airport_totals.head()
```

[35]:
```
    airport                                       airport_name  latitude  \
234     LAX                   Los Angeles International Airport   33.9425
220     JFK                  John F. Kennedy International Airport   40.6397
138     EWR                   Newark Liberty International Airport   40.6925
26      ATL  Hartsfield-Jackson Atlanta International Airport   33.6367
263     MCO                         Orlando International Airport   28.4294

     longitude  count
234  -118.4080  24794
220   -73.7789  24677
138   -74.1686  24405
26    -84.4281  23591
263   -81.3090  22308
```

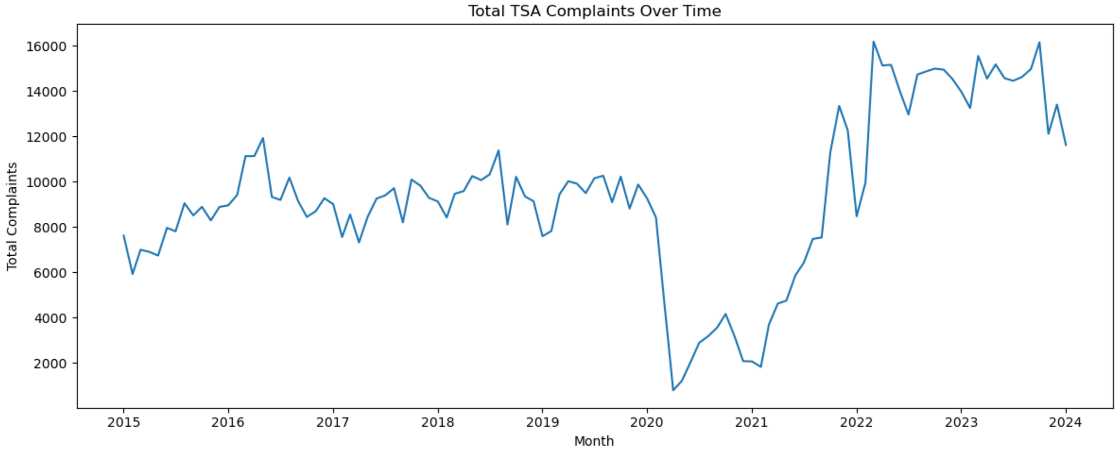
[36]:
```
top10 = airport_totals.head(10).copy()
top10["label"] = top10["airport"] + " - " + top10["airport_name"].
 ↪fillna("Unknown")

plt.figure(figsize=(12, 6))
plt.barh(top10["label"][::-1], top10["count"][::-1])
plt.title("Top 10 Airports by Total TSA Complaints")
plt.xlabel("Total Complaints (2015-2024)")
plt.tight_layout()
plt.show()
```



[24]:
```
top_cat10 = cat_totals.head(10).copy()

plt.figure(figsize=(12, 6))
plt.barh(top_cat10["clean_cat"][::-1], top_cat10["count"][::-1])
plt.title("Top Complaint Categories (Total)")
plt.xlabel("Total Complaints (2015-2024)")
plt.tight_layout()
plt.show()
```



[25]:
```
trend = (airport_df
         .groupby("ym")["count"]
         .sum()
         .reset_index()
         .dropna()
         .sort_values("ym"))

plt.figure(figsize=(12, 5))
plt.plot(trend["ym"], trend["count"])
plt.title("Total TSA Complaints Over Time")
plt.xlabel("Month")
plt.ylabel("Total Complaints")
plt.tight_layout()
plt.show()
```



[26]:
```python
top_cat = cat_totals.iloc[0]["clean_cat"]
sub_drill = (subcat_df[subcat_df["clean_cat"] == top_cat]
             .groupby("clean_subcat")["count"]
             .sum()
             .reset_index()
             .sort_values("count", ascending=False)
             .head(15))

plt.figure(figsize=(12, 6))
plt.barh(sub_drill["clean_subcat"][::-1], sub_drill["count"][::-1])
plt.title(f"Top Subcategories within: {top_cat}")
plt.xlabel("Total Complaints (2015-2024)")
plt.tight_layout()
plt.show()
```